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There's too many cars and not enough road – yeah, this is a pretty 'Captain Obvious' thing, but it has to be said before we move onto why it's mostly our fault. The solution: it's not about building more roads, but about people using less cars; relying more on public transit would be nice, though the number of cars is not really expected to drop in the nearby future.

People can't maintain a steady speed. The human inability to maintain a steady, constant speed on the road is responsible for a lot of congestion; this inability has been proven by Japanese researchers, when they asked people to drive constantly in a circle. Jams started appearing out of nowhere, people started braking eratically and everything went to hell.

"You're not driving into a traffic jam," says Vanderbilt. "A traffic jam is basically driving into you." He thinks autonomous cars will reduce this problem considerably. Self driving cars (which are already legalized in California) could work wonders here, and they could even communicate with each other, keeping a steady, constant distance between themselves – something which humans can't do, apparently.

We're also bad when it comes to doing the zipper merge. You're driving on one of the two lanes, when there's work in progress on one of them (or they just simply merge); humans are really bad, when it comes to doing this – especially Americans. American drivers typically merge into the right lane as soon as possible and form one long line, probably because they think it's bad manners to wait until the last moment or because they think they won't be allowed by other cars to merge late. In fact, says Vanderbilt, traffic would be much better if cars merged at the very end, like one big zipper. This would be safer, quicker, and would reduce road rage (which is continuously growing).

People don't want to use mass transportation

Vanderbilt's studies showed that if only 1% of all drivers would use mass transportation, the rest of them would reach their destination almot 20% faster (the 1% would get home faster as well). Even though they hate driving in traffic, even though it's more expensive and slow, people still want to drive their cars – this is called the 'car effect'.

"You can't just assume that as bad as traffic gets that people are automatically going to migrate to mass transportation," says Vanderbilt. Some people exhibit driver's courtesy, some people just take advantage. Self-driving cars could also greatly improve this aspect of driving. Studies have shown some remarkable characteristics of driving, like for example how older drivers are more likely to stop for others. People are more likely to violate traffic rules the closer they are to home — a "familiarity effect." Also, people tend to drive more closely to bicyclists when they are wearing helmets. Drivers of more expensive cars are more likely to honk in traffic, etc. All these are things we do almost unconsciously, but which all negatively contribute to traffic.

Executing the "zipper" merge. Road work often reduces two lanes of traffic down to one. In these situations, American drivers typically merge into the right lane as soon as possible and form one long line. The main reason they do this is because people think it's bad behavior to stay in the left lane and merge late.

http://www.zmescience.com/science/domestic-science/traffic-jam-sollution-30092013/

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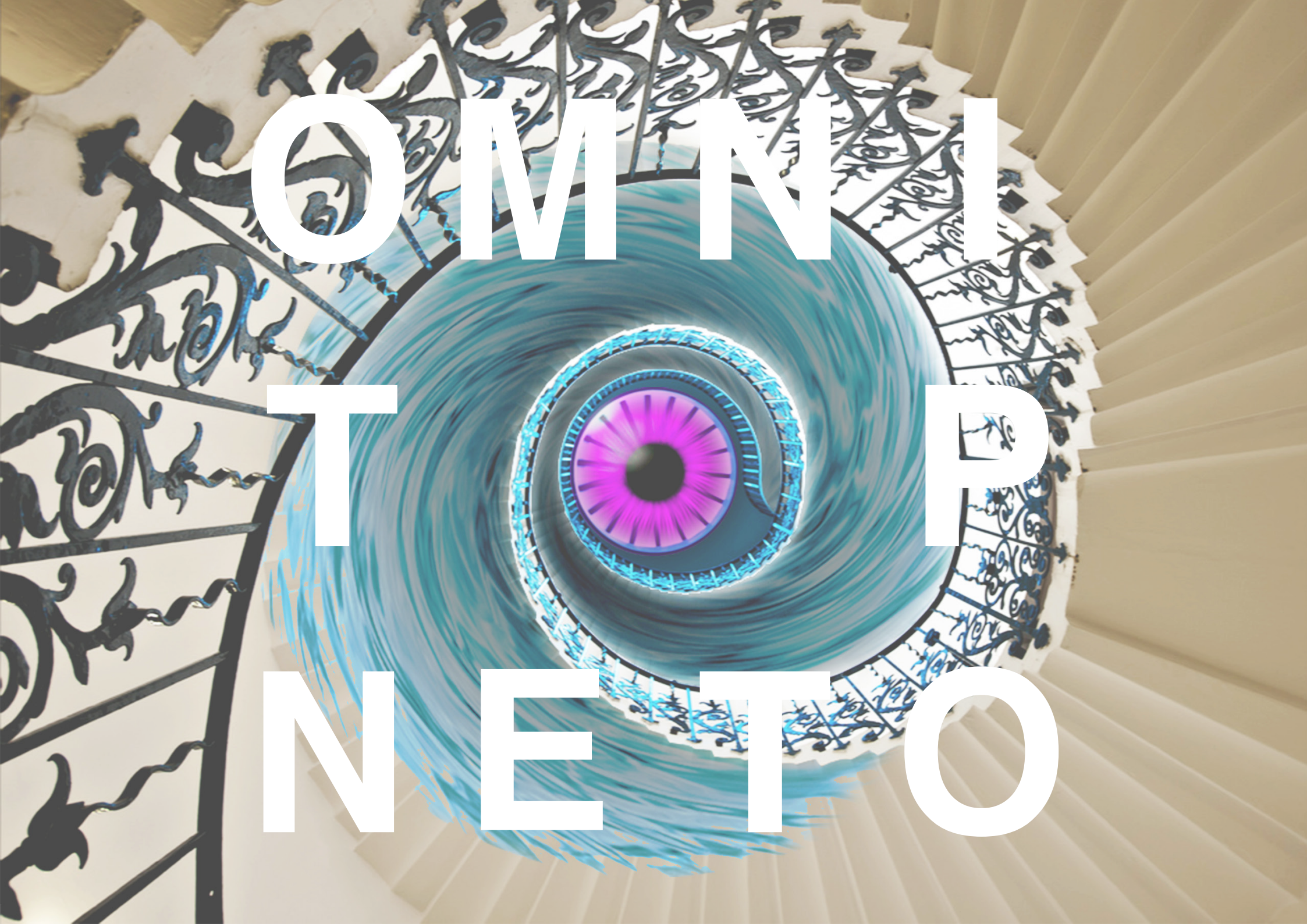
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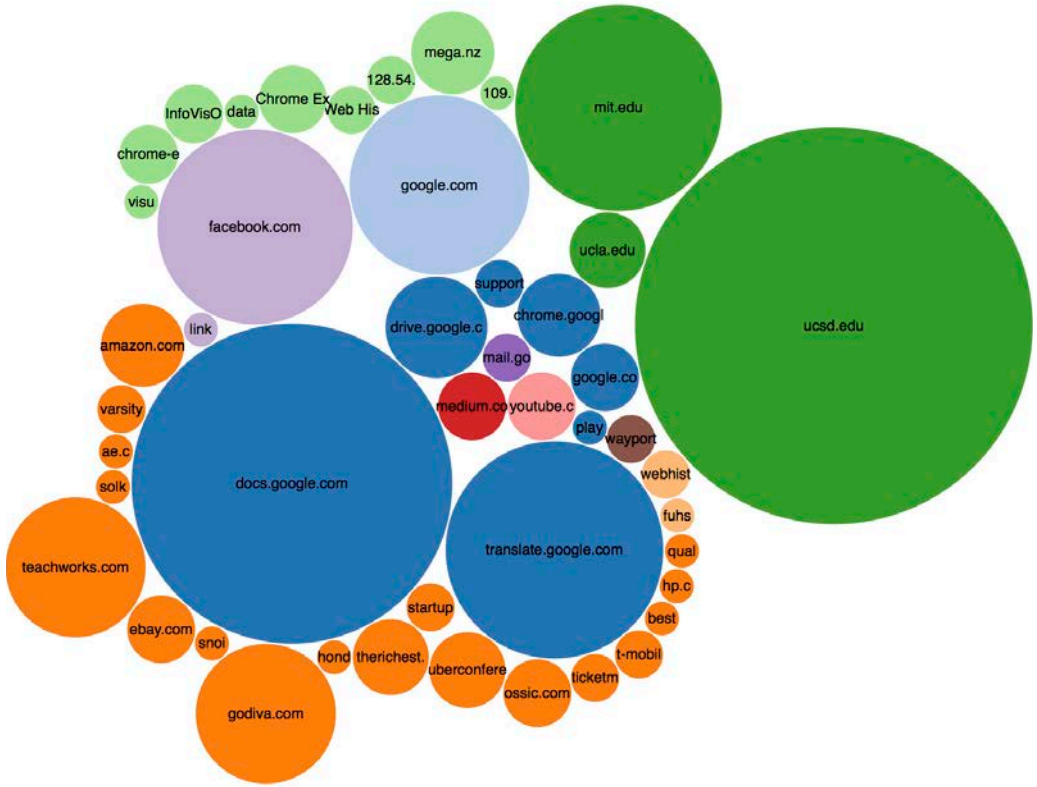

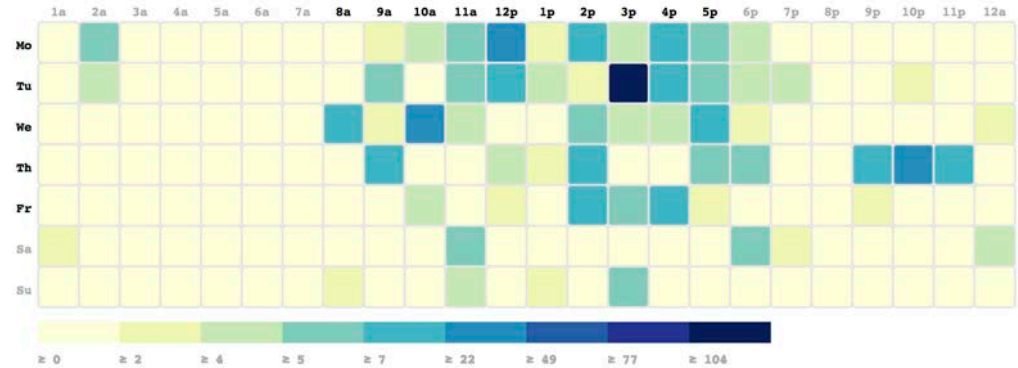

It's true, and has been for some time, that the Internet is connecting more people at a faster rate than ever before. Mobile devices have enabled more people to be online all the time.

We seemingly love the Internet. And in that love, some have a deep wish to be connected consistently (24/7, 365 days a year). The sector of tech loosely called "The Internet of Things" has become a piece of our culture that enables ordinary objects to be connected to the Internet – using sensors and controllers.

As Chet Pipkin, CEO and founder of Belkin, puts it: "The world is made up of trillions of things -- cars, planes, jet engines, exercise equipment, the items on my desk. And then there's the Internet. This category is about all of these things and the Internet, as we know it, coming together. Anything I can do over the Internet blended with my things."

"It's a brand new category, and it's getting formed in real time," he said.

Pipkin founded Belkin, a Los Angeles-based tech company, back in 1983.

Then, he could not have predicted that we would one day have an Internet, much less be wearing it. What he did predict correctly is that being a conduit between people and the things they love to do would be a great place to plant a company for the long-term.

Belkin started in consumer electronics in the 1980s. Today they have become a leader in "The Internet of Things" space. A sector that Pipkin not only wants to master, but lead all together on a global level.

"We will bring our own devices to the market that will lead the Internet of Things -- creating products that take physical things and give them an on-ramp onto the Internet." Belkin's first "Internet of Things" system was WeMo, a simply Wi-Fi-based home automation network. WeMo might automatically cool or heat your home, turn on and off lights, and otherwise set up one's home for automatic control (i.e. you could change or set controls for your garage door, or turn on the porch lights -- from your smartphone or any computer).

As everything becomes wired, where does that leave us?

With greater energy consumption becoming a concern, Belkin has thought ahead and made energy a key focus of their company – both in how people can monitor their own energy use, and figuring out what would actually drive people to want to conserve.

"We firmly believe that we are fortunate that we are able to have a lot of the world's resources available to us. If we look at the rate we consume resources per capita -- it's staggering. So, what can we do to more prudently and more judiciously use the resources on our Earth?"

Their technology called Echo allows one to conserve water and electricity. "By monitoring consumption levels, users can see it and feel a reward for conserving. It allows me to monitor, measure, and manage all the things in my life in a really approachable, fun way."

Echo, at its core, is an advanced technology system that uses a single, low-cost sensor in each home and signal-processing algorithms – to show people how water, electricity, and natural gas are each being used. The intent is to identify waste and recommend ways to save. The wearables (i.e. smart watches, bands that monitor heart rate and steps) market is also on Belkin's radar, though they have not yet publicly announced a product in the space. Pipkin feels strongly about the category and hinted at products in the wearables space being a huge part of the Internet of Things all together.

What then of "unplugged" time or down time from the Internet? (Something that even Internet junkies have been saying that humans need, in order to digest all the information we are exposed to).

Chet says his unplugged moments are more valuable than his plugged moments.

However, the technology his company has created has allowed for less worry while he is unplugged (i.e. one example might be - receiving automatic notifications in the case of an emergency).

Whether you are someone who is jacked into the Internet all day long or someone who still reads by candlelight – one reality of our modern life is that we are experiencing the Internet in different forms at all times – enjoying access to more data, information, thoughts, and ideas than ever before.

https://www.forbes.com/sites/lorikozlowski/2014/04/23/everything-is-connected-what-the-internet-of-things-means-now/2/

[privacy]

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Apple Watch

Bluetooth privacy protections —or lack thereof —were central the study's findings. Of the eight devices tested, Apple's wearable was the only one which regularly altered the MAC address broadcast by its Bluetooth radio. Randomization of the MAC address on Bluetooth Low Energy products is accomplished by a BLE feature known as "LE Privacy."

This is important, because unpaired Bluetooth products are designed to send "advertising" packets at regular intervals for discovery —that's how your iPhone knows that there's a nearby Apple Watch available for pairing. Without this feature, researchers at Canadian privacy non-profit Open Effect and the University of Toronto note that it's relatively trivial to track the movements of individual users when their fitness bands are not actively paired with a device.

Fitbit blamed the "fragmented Android ecosystem" for the lack of LE Privacy support.

Contacted by the researchers about the fault, Fitbit noted that compatibility issues within the "fragmented Android ecosystem" prevent them from adding LE Privacy, despite hardware support in their products. Through corporate parent Intel, Basis noted that using the Peak while not paired to a smartphone was an edge case and did not commit to a fix. None of the other companies in the test —Garmin, Jawbone, Mio, Withings, or Xiaomi —came back with "notable responses."

In addition to the Bluetooth issues, several companion software packages were found to be insecure. The researchers were variously able to intercept and read fitness data or write false data to disk.

The Garmin Connect app does not use HTTPs for connections, allowing a man-in-the-middle attack to read and write data. A similar attack was possible against Withings's Health Mate app on Android, while Jawbone's Up could allow users to send arbitrary fitness data to the cloud, an issue with potentially severe consequences:

"These findings concerning fitness tracker data integrity could call into question several real-world uses of fitness data," the researchers wrote. "Fitness tracking data has been introduced as evidence in court cases...meaning that at least some attorneys are relying upon generated fitness data as a possibly objective indicator of a person's activities at a given point in time. For Jawbone and Withings we created fraudulent fitness data which indicated that a passive measuring device, the fitness device, recorded a person taking steps at a specific time when no such steps occurred."

http://appleinsider.com/articles/16/02/08/apple-watch-comes-out-ahead-in-study-of-fitness-tracker-privacy-security

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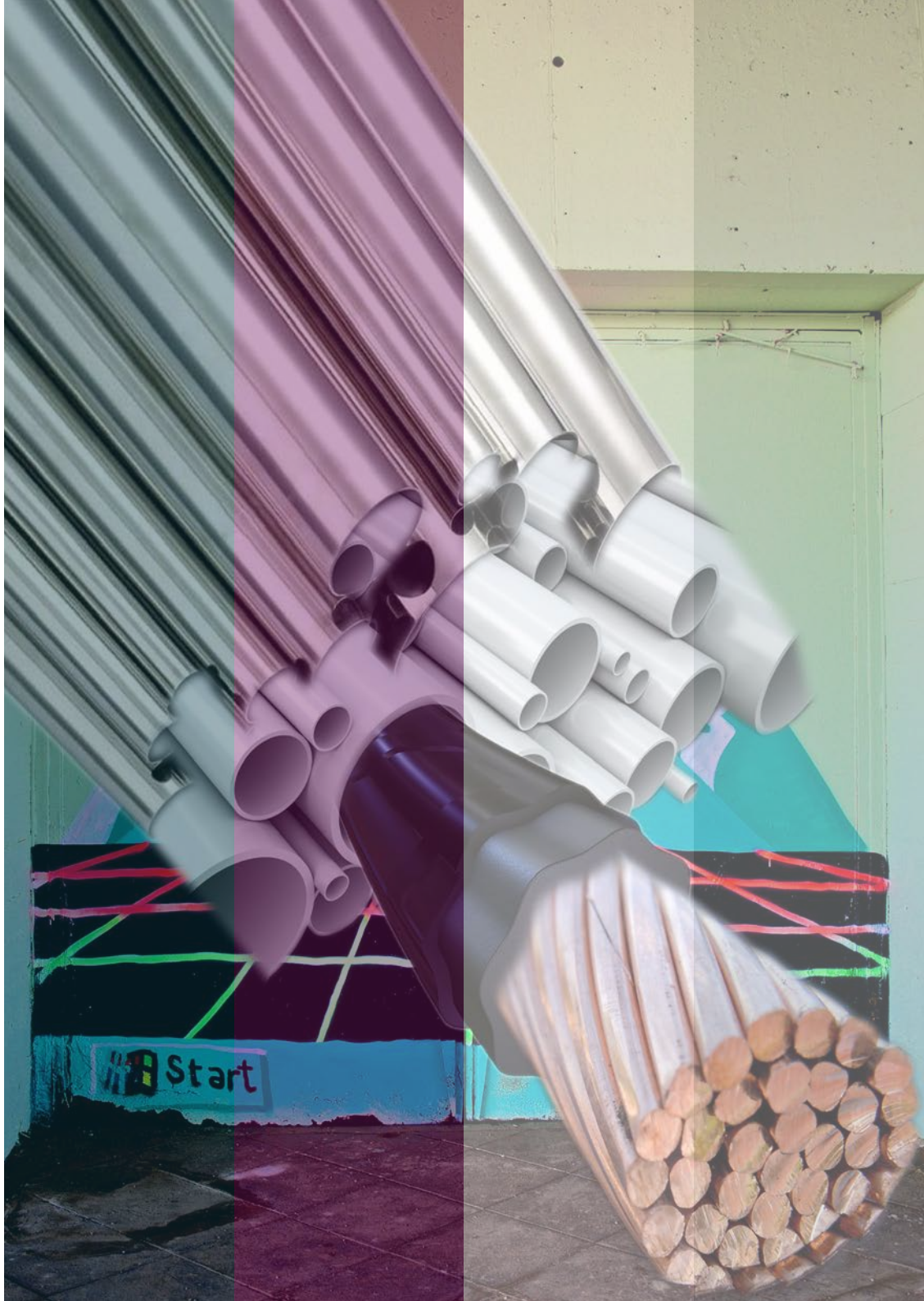

INDUSTRIAL

It all starts with a rusty heap of scrap metal. Anything containing copper will do... because any extra casings burn off when it gets melted down into copper soup.

To get temperatures up to boiling, factory workers inject oxygen under extremely high pressures.

The molten copper (which still contains quite a few impurities) starts flowing into molds.

The molds are then dipped in water to cool down to manageable temperatures.

At which point it gets stretched and stretched until it's thin enough to meet the necessary diameter.

The cooled molds get sent off to shed their impurities in giant electrolysis baths.

The pure, molten copper is cooled and shaped into a large snake.

And out come much prettier copper sheets.

To be made into wire, this copper gets sent out to a new shop to be melted down once again.

http://gizmodo.com/the-incredible-fiery-process-of-making-copper-wire-1559282588

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UNDER COVER

The fury of the great storm Sandy shocked a lot of people, like John Miksad, vice president of the New York electric utility Consolidated Edison. “We hit 14-foot tides — that was the biggest surprise,” he told a press conference this week. “The water just kept rising and rising and rising.”
That rising water flooded streets, buildings and parts of the city’s underground electricity grid. Hundreds of thousands of New Yorkers lost power. But it might have been worse if the power lines had not been underground. Manhattan’s grid sits like a gigantic plate of spaghetti under the city’s streets and buildings — miles and miles of electric cable. After the 14-foot tide went up and over, it went down under,

One reason Manhattan’s underground system is resilient is that it gets wet all the time. That’s according to energy expert Roger Anderson at Columbia University, who has been a consultant to Con Ed. “In the winter we put salt down on the roads,” he explains. “The salt melts, and the transformers under the streets are routinely dealing with saltwater. So all that Con Edison needs to do is to drain the water out of the transformers, hose them down, and flip the circuit breaker again.”
Anderson concedes, though, that flooding poses other threats. “Flooding has all kinds of contaminants in the water, a lot of oils, and also it comes in violently — it can wash away things.” Things like transformers that can get knocked off their moorings.

NEW YORK

Even as the grid gets fixed, there’s the problem of the subways. Several subway tunnels have been flooded, and there’s lots of electrical equipment in those tunnels. For example, if you’ve been on a New York subway, you’ve seen the red and green lights along the tunnel. Those are switches that monitor where every train is, and they all have to be checked. “And in order to do that,” Anderson explains, “you’ve got to physically walk or ride the length of the track.”

It’s a big, big job with a big price tag. So, it’s to be expected that New Yorkers are asking: Will climate change make this keep happening over and over? It’s the second year in a row that a hurricane has hit New York. Last year it was Irene.

soaking a lot of that spaghetti network — cables, transformers, switches; all the stuff that’s hidden under those manhole covers in New York’s streets.

“It’s a whole different world down there, believe me,” says Gregory Reed. He teaches power engineering at the University of Pittsburgh, but he used to crawl down those manholes for Con Ed, to fix their grid. “It is very cramped,” he says. “There’s seldom room for more than one person at a time to do their work. So it’s very difficult conditions.”

And that’s what a small army of workers is doing now — squirming through narrow tunnels under Manhattan. They’re checking hundreds of transformers that regulate the electricity’s voltage — those gray cylinders you normally see perched on power poles. And they’re looking for cables that got damaged. “It’s not always easy to predict where that fault is in the underground system,” says Reed, “because we can’t see it, so what you have to do is go out and test it section by section.” Even equipment that’s not obviously damaged has to be tested. And that could mean turning power off and on for a while in a neighborhood; that’s pretty annoying if you just got your power back. But Reed says it would’ve been worse if the city’s grid were aboveground. “Consider this case,” he says. “If all of the infrastructure was overhead, the vast majority would be laying on the ground.” In fact, Con Ed officials say repairing aboveground regions of the area will take three or four times as long as fixing the underground network.

Climate scientists caution that no single weather event is proof of climate change.

But computer projections of future warming do predict bigger storms.

At a press conference in New York, Mayor Michael Bloomberg said he’s seen enough to act. “Let’s assume that we decide that we’re not damaging our planet and later on find that we were,” the mayor said. “It literally could be too late.”

But just what to do to protect New York awaits months of post-storm analysis by engineers and oceanographers and other experts, followed by action by politicians and industry, who will have to raise many billions of dollars to pay to keep the Big Apple afloat.

http://www.npr.org/2012/11/02/164109929/fixing-nyes-underground-power-grid-is-not-easy-task